

Potential FPI scenario examples



IOGP has set a course for our industry to focus on preventing fatalities and life-changing injuries.

IOGP Fatality and Permanent Impairment (FPI) injury definitions are designed for use in energy industry reporting. The full definition, along with examples and FAQ can be found at <https://iogp.org/fpi>.

Within the FPI reporting methodology the following definitions are used:

- **Actual FPI:** An incident that results in a fatality or permanent impairment
- **Potential FPI:** An incident that results in an injury (first aid and recordable) that could have realistically resulted in a fatality or permanent impairment
- **Near Miss FPI:** An event that could have realistically resulted in a fatality or permanent impairment where there was no actual injury

Note: IOGP defines a High Potential Event as any incident or near miss that could have realistically resulted in one or more fatalities. An actual, potential or near miss FPI may or may not be a high potential event.

In order to help organizations categorise incidents and events in terms of potential FPI, a list of scenario examples has been produced. Each scenario is described briefly and grouped by activity, cause, causal factor and consequence. The potential outcome (PI or Fatality) is indicated against each scenario, and a comments column provides additional information and examples. The list is not intended to include all possible scenarios, and only relates to potential FPI (i.e. those where there was an injury).

For the complete definitions, examples and FAQ please visit www.iogp.org/fpi



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Categorization examples for potential Fatalities or Permanent Impairment injuries (FPI) with hurt, as a result of an acute, single incident

Category		Potential outcome	Comments - how to categorise
ACTIVITIES*			
* For an actual/potential injury involving a person being caught in under or between or struck by an object please refer to the respective definition below. This will provide the considerations applicable to determine whether the realistic worst outcome would be a fatality or permanent impairment injury.			
Diving (incl. decompression), subsea, ROV	Diving related		
	Decompression sickness - Central Nervous System (CNS)	Fatality / PI	Fatality if untreated. Length of exposure will determine outcome and damage caused by the gas bubble can lead to permanent disability.
	Improper gas mixture	Fatality / PI	Fatality if untreated. Length of exposure will determine outcome.
Excavation, trenching, ground disturbance	Excavation		Reference IOGP Start Work Checks for excavation, energy isolation and confined space entry hazard controls.
	Wall collapse or material/soil fall into excavation	Fatality	Assumes excavation with person inside, or planned to have person inside. Consider depth of excavation, excavation material, soil type, water accumulation/infiltration, and worker position for risk of engulfment. See Dropped or Falling Objects for dropped materials. Consider also other hazards such as dropped objects, falls from height, etc.
	Equipment striking an unknown underground utility or unexploded ordnance (UXO)	Fatality	Consider type and state of utility (service, operating pressure, etc.) and type of equipment used in strike. See Electrical for underground electrical utility strike. See Other Energy Sources / Residual Product for underground pipeline strike. For UXO, consider the realistic potential for detonation of the device discovered.
	Heavy machinery, staged equipment, or excavated material located close to the excavation edge	PI	For equipment operator working within or planned to be within the excavation, rate as potential Fatality. Consider established guidelines for determining proximity expectations, including in staged areas, for escalation to potential Fatality.
	Other potential situations / conditions	PI	Untested/unsafe atmospheres (see confined space); Fall potential (see falls from height); Workers near excavation equipment (see transport - work around mobile equipment); Lifts performed by excavators or mechanical lifting equipment (see lifting & hoisting).
Lifting, crane, rigging, deck operations	Lifting & hoisting		Reference IOGP Start Work Checks for safe mechanical lifting and man-riding hazard controls.
	Standing underneath lifted load	Fatality	
	Unexpected movement of lifted load or failed equipment when personnel are in the line of fire	Fatality	This could include man riding activities.
	Contact with a lifted load in motion with any part of the body	PI	For example, guiding a load with hands during any motion except for the final setting of the load (i.e. guiding a pump or motor onto a baseplate / anchor bolts for the last ~1' of vertical motion), or tagline entanglement as taglines are considered part of the lifted load.
	Standing between lifted load in motion and stationary object or within swing radius of lifting appliance including counter weights	Fatality	Individuals standing in the line of fire or in a crush point.
Transport	Transport - land (See definition below for Motor Vehicle)		Workforce injury only. Reference IOGP Start Work Check for driving hazard controls.
	Motor vehicle crash resulting in rollover beyond its side (>90 degree)	Fatality	
	Motor vehicle crash resulting in rollover to its side only (≤90 degree)	PI	If seat belts are not worn then Fatality. If fatigue or (micro)sleep involved then Fatality.

Category		Potential outcome	Comments - how to categorise
ACTIVITIES*			
Transport	Motor vehicle crash at speed of ≥ 50 mph (80 kph)	Fatality	Includes collision with a fixed object or large animal at the speed listed, and/or collision between two or more vehicles involving a speed differential of vehicles (including motorcycles) traveling in same or different directions.
	Motor vehicle crash at speed of 30-50 mph (48-80 kph)	PI	Front, Side & Rear collisions. If no seat belts, then Fatality. Potential for increased severity if occupant on side of impact. Includes collision with a fixed object or large animal at the speed listed, and/or collision between two or more vehicles involving a speed differential of vehicles travelling in different directions. Consideration should be given to the proper wearing of 3 point retractable seatbelts and the safety rating of the vehicle (e.g. NCAP rating). If the vehicle involved has a 5 star NCAP safety rating, in the case of minor hurt only (e.g., first aid, medical treatment only) this may not be a PI. If fatigue or (micro)sleep involved then Fatality.
	Motor vehicle crash at 15-29 mph (24-47 kph) speed without seat belt	PI	
	Pedestrian /cyclist involvement in MVC or work around mobile equipment (WAME)		Workforce injury only. Reference IOGP Start Work Check work around mobile equipment hazard controls.
	Person impacted by moving mobile equipment or vehicle at speed of ≥20 mph (32 kph).	Fatality	If pedestrian/cyclist struck by vehicle or mobile equipment, pedestrian chances of survival are only reasonably high in collisions at low collision speeds (lower than 20 mph (32 kph))**. In all of the pedestrian datasets, the risk of fatality increases slowly until impact speeds of around 20 mph (32 kph). Above this speed, risk increases rapidly***.
	Person impacted by moving mobile equipment or vehicle at speed of 10-20 mph (16-32 kph).	PI	If pedestrian/cyclist struck by vehicle or mobile equipment, potential severity increases if reasonable for person to be 'caught between' or 'run over'.
	Person impacted within the swing radius/line of fire of the (mobile) equipment or vehicle or in their blind spot	Fatality / PI	Line of fire should be considered if the equipment is being operated or has the potential to be operated (operator in the cabin) in the direction of the person (e.g., pedestrian, cyclist). Consideration should be given to the actual swing radius and / or the "blind spots" of the specific equipment.
	Person impacted by uncontrolled movement of mobile equipment	Fatality / PI	Uncontrolled movement of mobile equipment. Examples: parking brake not applied, unchocked trailer on a slope, parked on ice or a slippery surface.
	Transport - air		
	Aircraft event	Fatality / PI	Any aircraft accident or event involving manned fixed or rotary wing aircraft. Examples: uncontrolled landing on land or water, unplanned or uncontrolled people movement in the aircraft operating area, contact with moving parts of the aircraft on the helideck or apron.
	Transport - water, incl. marine activity		Reference IOGP Start Work Check Work near water hazard controls.
	Water related event	Fatality / PI	Incident or event in which water played a significant role, e.g. involving vessels, equipment or boats designed for transporting people and goods over water (including inland, marine, ice roads and marsh/swamp), supply vessels, crew boats, personnel transfers over water, lifeboat launching in emergency or drill scenario, person impacted by break of tensioned mooring lines, drowning.
	Rail transport		
	Rail transport event	Fatality / PI	The derailment or collision of a train with any other train, object, vehicle or person.

Category		Potential outcome	Comments - how to categorise
CAUSES*			
* For an actual/potential injury involving a person being caught in under or between or struck by an object please refer to the respective definition below. This will provide the considerations applicable to determine whether the realistic worst outcome would be a fatality or permanent impairment injury.			
Assault or violent act	Assault or violent act		
	Security or workplace violence event	Fatality / PI	Consider the body part affected. Examples: assault involving a weapon of extreme force, civil unrest, security breach, terrorism, insurgency, piracy, violent crimes, kidnap or hostage taking, attack by pirates, explosive attacks.
Caught in, under or between	Caught in, under or between		
	Caught in, under or between	Fatality / PI	Injury where injured person is crushed or similarly injured between machinery moving parts or other objects, caught between rolling tubulars or objects being moved, crushed between a ship and a dock, or similar incidents. Also includes vehicle incidents involving a rollover. Includes entanglement such as the potential for a person to be pulled into energised/ moving equipment.
Confined space	Confined space		Reference IOGP Start Work Check for Confined space entry hazard controls.
	Unauthorized confined space entry (CSE) without authorized gas testing	Fatality	In a permit controlled confined space.
	Authorized / permitted CSE where ventilation contaminated by an outside source or an ignition source is introduced from outside source	Fatality	In a permit controlled confined space.
	Gap or failure in required CSE hazard controls for authorized / permitted CSE	PI	Examples: Atmosphere not monitored as required by the permit, or prior to or during entry, or ventilation failure/ unauthorized change during entry, competent CSE Attendant (hole watch) not in place during entry, effective rescue plan not in place or rescue team not available, malfunction of breathing apparatus during a confined space entry that could have resulted in the loss of breathable air, working in a confined space with hazardous atmosphere with no breathing apparatus, an incident occurring in a confined space which results in an emergency rescue from the confined space.
Dropped objects	Dropped or falling objects	Drops Calculator	Drops Calculator should be used to determine Worst Case Potential Hurt Level. Reference IOGP Start Work Check for rig floor tubular handling, safe mechanical lifting, or work at height hazard controls.
	Person impacted by dropped or falling object	Fatality	Using the drops calculator include red drops as a Fatality.
		PI	Using the drops calculator include orange drops as a PI.
Electrical	Exposure: electrical		Reference IOGP Start Work Checks for energy isolation, de-isolation and re-energizing and energized/live electrical systems hazard controls.
	Unprotected exposure to energised equipment ≥ 110 Volts	Fatality	Contact with a current carrying component (regardless of site conditions, e.g. wet/dry/clean). Examples: digging up cable with excavator, cutting into a wall and contacting cable, cutting an energized cable, contact with overhead power lines.
	Any potential exposure for arc flash during work activity without arc flash hazard PPE within the arc flash hazard boundary	PI	Assumes the event was such that an arc flash occurred, or could have readily occurred, while a worker was within the arc flash hazard boundary or qualified worker boundary.
Explosions or burns	Explosions or burns		
	Fire and explosion	Fatality / PI	Flash fire, process fire, equipment fire, lithium-ion battery fire. See consequence section for specific criteria for burns.
	Hot work		Reference IOGP Start Work Check for Hot work hazard controls.
	Hot Work performed in the immediate vicinity of a flammable or explosive atmosphere	Fatality	For example: without adequate isolation, without removal of residual product, without all permit controls in place, or no gas testing performed to determine if a hazard atmosphere is present. Uncontrolled source of ignition in a flammable or explosive atmosphere. Potential for fire/explosion.

Category		Potential outcome	Comments - how to categorise
CAUSES*			
	Hot Work performed outside of a controlled area with potential for fire	PI	Ignition sources are not identified and controlled. Flammable/combustible materials have not been removed or isolated.
Falls from height	Slips, trips & falls from height (people)		Includes falls through holes or openings in the walking surface to a level below. Reference IOGP Start Work Checks for work at height and man-riding hazard controls.
	Fall greater than 10 feet (3.0m)	Fatality	Assumes fall to solid surface/object or soft surface (mud, water). Including asset integrity failure, e.g. handrail or grating. Suspension trauma should also be considered in this category. Landings below 15 feet (4.5 metres) could be rated as PI.
	Fall between 6 feet - 10 feet (1.8 - 3.0 metres)	PI	Assumes fall to solid surface/object. Hitting obstacles during fall may elevate to a rating of Fatality. Permanent guarding such as ladder cage may decrease severity.
	Fall down 6 feet (1.8 metres) or more of stairs	PI	Assumes fall down stairs/steps of a normal slope. Inability to 'manage' fall could result in debilitating blow to head.
Struck by	Struck by		
	Struck by	Fatality / PI	Incidents or events where injury results from being hit by moving equipment and machinery, or by flying or falling objects. Also includes vehicle incidents where the vehicle is struck by or struck against another object. Consider the body part affected. Any event that involves an injury to the head or torso may be considered to have Fatality potential. Events that involve injuries to extremity body parts may be considered potential PI.

Category		Potential outcome	Comments - how to categorise
CAUSAL FACTORS*			
* For an actual/potential injury involving a person being caught in under or between or struck by an object please refer to the respective definition. This will provide the considerations applicable to determine whether the realistic worst outcome would be a fatality or permanent impairment injury.			
Bypassing safety controls	Safety devices		Reference IOGP Life-Saving Rule Bypassing safety controls.
	Disabling or bypassing safety controls without authorization and alternative controls in place	Fatality	For example, fire safety/suppression systems in occupied building, mobile equipment backup alarm, disabling crane load limit or anti-two block devices, removal of hand rails, etc.
	Disabling or bypassing a guard on machines or equipment without authorization and alternative controls in place	Fatality	Focus is primarily on machine and equipment guarding, not tool guards. If work as executed exposed worker to potentially fatal sources of energy, refer back to primary category, e.g., electrical and energy isolation.
	Disabling or bypassing a guard on a tool without authorization and alternative controls in place	PI	Work as executed exposed a worker to potential sources of energy that could cause life altering injuries. If the energy source is electrical or from rotating components, refer back to primary category (e.g., electrical, or other energy sources).
Energy Isolation	Other energy sources / residual product		Reference IOGP Process Safety Fundamentals and Start Work Checks Energy isolation, and de-isolation and re-energizing hazard controls.
	Unprotected contact with rotating equipment, uncontrolled maintenance on live equipment or unguarded powered equipment with known pinch points	Fatality / PI	Consider the body part affected. Any event that involves an injury to the head or torso, or the potential for a person to be pulled into energised/moving equipment, may be considered to have Fatality potential. Events that involve injuries to extremity body parts may be considered potential PI.
	Unexpected release of stored mechanical energy (tension release / pressure release, pressure relief projectiles, whipping tubing / pipe, etc.)	Fatality / PI	Consider the pressure, mass of the object, velocity of object, and body positioning during unexpected release. Not a FPI if potential is for a simple broken bone. PI for possible loss of body part (see FPI definitions), eye hearing loss, compound fractures, or other PI consequence. Increase to Fatality for fatal crush or fatal contact potential.

Category		Potential outcome	Comments - how to categorise
CAUSAL FACTORS*			
Energy Isolation	Fire and explosion	Fatality	Flash fire, process fire, equipment fire.
	Gap or failure in safe isolation controls	Fatality / PI	Failure / inappropriate isolation, including electrical isolation. Wrong line opened / cut.
	Other potential situations / conditions	Fatality / PI	Unplanned movement during equipment installation and/or decommissioning.
	Process releases		Reference IOGP Process Safety Fundamentals and Start Work Checks Energy isolation, and de-isolation and re-energizing hazard controls.
	Loss of primary containment	Fatality / PI	Consider the release rate, volume, material released and proximity of people to the release site, affected area and any potential sources of ignition. Example: flammable/toxic release with potential exposure of people that could result in fatality or PI.
	Breaking containment / opening process equipment (product exposure)	Fatality / PI	Consideration includes exposure to residual product, volume and pressure, e.g. temperature, toxic vapor, acids, auto-ignition of product, etc.

Category		Potential outcome	Comments - how to categorise
CONSEQUENCE			
Loss of body part	Amputation-related		
	Hand, foot, limb	Fatality	Any event without local emergency response capability.
	Finger, thumb, toe	PI	Any event resulting in losing part of the finger, thumb or toe beyond the tip, including a piece of bone but not beyond the first joint, or where the amputated digit was successfully reattached with return of functionality.
Reduction of organ function	Eye-related		
	Loss/reduction of eyesight	PI	Any event where physical or chemical eye trauma occurred or had reasonable potential to occur. For foreign bodies in the eye consider the nature of the foreign body, and the ability to treat/remove it. Exposure to corrosive chemicals, strong alkalis or acids, or other severely irritant substances as per chemical classification on safety data sheet.
	Hearing-related		
	Loss/reduction of hearing	PI	Any event with potential for permanent loss or reduction of hearing, including resulting in persistent tinnitus.
	Exposures		
	Anaphylactic reactions	Fatality	Health professional confirmed.
	Unprotected/uncontrolled exposure above immediately dangerous to life or health (IDLH) level	Fatality	Example: asphyxiating or toxic atmosphere, e.g. H2S exposure >100 ppm, exposure to oxygen depleted atmosphere, CO exposure.
	Unprotected/uncontrolled ingestion, injection or contact with toxic materials	Fatality	Example: single chemical exposure above toxicity level by means of entry into the body as per safety data sheet. Example: exposure to hazardous flora or fauna (including animal attack or insect bite/sting), e.g. exposure to poisonous plants or venom.
	Exposure to corrosive chemicals, strong alkalis or acids, or other severely irritant substances as per chemical classification on safety data sheet.	PI	Example: exposure to hydrofluoric acid.

Category		Potential outcome	Comments - how to categorise
CONSEQUENCE			
Reduction in psychological, social or cognitive function	Reduction in psychological, social or cognitive function		
	Coma	Fatality	Any event resulting in coma with Glasgow Coma Scale (GCS) < 9.
	Concussion/compression/brain injury	PI	Health professional confirmed - ongoing impact to emotional regulation, speech, memory, or other cognitive dysfunction. Any event resulting in a moderate traumatic brain injury (TBI) with GCS 9-12.
	Head injury	Fatality / PI	If the individual has a minor head injury (e.g. a glancing blow from a falling object) or impact just missed the individual's head (within a few inches) could be elevated to Fatality. For example if the object had been a few inches different in position and had hit them fully on the head.
	Post injury psychiatric disorders	PI	Health professional confirmed - including Post Traumatic Stress Disorder (PTSD) with inability to return to previous work role and/or schedule.
Reduction in skin and musculoskeletal function	Reduction in skin and musculoskeletal function		
	Fire or burn (chemical or thermal)	Fatality / PI	Any event involving fire or burn with the potential to cause permanent skin function impairment/reduced mobility. May be potential Fatality depending on extent of burns.

** Source: SWOV (Institute for road Safety research) <https://swov.nl/en/fact/pedestrians-what-correlation-between-collision-speed-cars-and-pedestrian-fatality-rate>

*** Source: Transport Research Laboratory https://nacto.org/docs/usdg/relationship_between_speed_risk_fatal_injury_pedestrians_and_car_occupants_richards.pdf

Definitions

Potential FPI: An incident that results in an injury (first aid and recordable) that could have realistically resulted in a fatality or permanent impairment.

Near Miss FPI: An event that could have realistically resulted in a fatality or permanent impairment where there was no actual injury.

Motor Vehicle: Any mechanically powered vehicle used to transport people or property, including any load on or attached to the vehicle (e.g., a trailer). This includes motorcycles. Specifically excluded from the definition of motor vehicle are vehicles operated on fixed rails and onsite vehicles that are not capable of more than 10 mph (16 kph).

Vehicles are split into four sub-categories:

- Heavy Vehicle (HV)
- Light Vehicle (LV)
- Mobile equipment
- Motorcycles